ISLAMIC UNIVERSITY OF TECHNOLOGY
Department of Computer Science and Engineering (CSE)
Course Outline and Course Plan

Name of the Teacher	Samia Nawsheen	**Position**	Junior Lecturer
Department	CSE	**Programme**	B.Sc. in SWE
Course Code	SWE 4201	**Course Title**	Object Oriented Concepts I
Academic Year	2024-25	**Semester**	Summer
Contact Hours	3.0	**Credit Hours**	3.0
Text books and Reference books	1. The Object-Oriented Thought Process - Matt Weisfeld 2. Head First Object-Oriented Analysis and Design - Brett McLaughlin, Gary Pollice, David West		
Prerequisites (If any)	1. CSE 4107: Structured Programming 2. SWE 4101: Introduction to Software Engineering		

Teaching Methods/ Approaches	☐ **Lecture√**	☐ **Group discussion**	☐ **Demonstration**	☐ **Problem solving√**
	☐ **Project√**	☐ **Others: Tutorial classes√**		
Teaching aids	**Multi-media √**	**OHP**	**Board and Marker√**	**Others**

Course Assessment Method						
Attendance (0%)	**Quiz 20% of Total Marks (Best 3 out of 4)**				**Mid Semester (40%)**	**Semester Final (40%)**
	1st Quiz	**2nd Quiz**	**3rd Quiz**	**4th Quiz**	As per IUT schedule	As per IUT schedule
	3rd Week	6th Week	10th Week	13th Week		

Grading Policy					
Marks out of 100	**Letter Grade**	**Grade Point**	**Marks out of 100**	**Letter Grade**	**Grade Point**
80 - 100	A+	4.00	55 - 59	B-	2.75
75 - 79	A	3.75	50 - 54	C+	2.50
70 - 74	A-	3.50	45 - 49	C	2.25
65 - 69	B+	3.25	40 – 44	D	2.00
60 - 64	B	3.00	00 - 39	F	0.00

Course Contents
Introduction to Object Oriented Concepts: class, object, encapsulation, inheritance, interfaces, Using UML to model a Class Diagram, Constructor, Polymorphism, Aggregation and Composition, Error handling and The concept of scope; The Anatomy of a Class – The Name, Comments, Attributes, methods, Constructors, Accessors, Modeling Real World Systems, Designing with Reuse, Extensibility, Maintainability in Mind; Programming lessons - Introduction to Java – Java Virtual Machine (JVM) and Java Runtime (JRE), Java Development Kit (JDK), Integrated Development Environment (IDE) for Java, Writing programs in java and learning java syntax, features and libraries.

Course Objectives
The subject aims to equip the students such that they will be able to understand fundamental concepts of Object Oriented Programming (OOP) and do the following: 1) Develop a foundational understanding of Object-Oriented Programming (OOP) concepts. 2) Build the ability to analyze problems and identify suitable object-oriented solutions. 3) Enable students to design and implement software solutions using object-oriented principles and programming languages. 4) Develop the ability to evaluate and improve software design using concepts such as coupling and cohesion.

Mapping with CO, PO and Bloom's Taxonomy			
CO No.	**Course Outcomes (CO) Statement**	**Levels of Bloom's Taxonomy**	**Matching with Program Outcome (PO)**
CO1	Explain fundamental OOP concepts	C2	PO1
CO2	Analyze problems and identify appropriate object-oriented approaches	C4	PO2
CO3	Design and implement solutions object-oriented programming principles	C3	PO3, PO5
CO4	Evaluate software design using concepts like coupling, cohesion, and relationships	C5	PO4

Weekly plan for course content and mapping with CO		
Weeks	**Topics**	**COs**
1	Introduction, Procedural programming vs OOP, Features of OOP, Concept of Classes and Objects	CO1
2	Java Basics, Classes and Objects in Java	CO1, CO3
3	Constructors, Variables, Data types	CO3
4	Java operators, Loops, Conditional Statements, Arrays	CO3
5	Encapsulation, Getters, Setters, Scope, Lifetime, Access Modifiers	CO1, CO3
6	Compile-time Polymorphism (Method Overloading)	CO1, CO3
7	Inheritance	CO1, CO3
Mid-Semester Examinations		
8	Runtime Polymorphism (Method Overriding), Superclass Reference, Typecasting	CO1, CO3

9	Abstraction, Abstract Classes, Interfaces, Subtyping	CO1, CO3
10	UML Class Diagram, Relationships (Aggregation, Composition, Delegation)	CO3
11	Arraylist, Generics	CO3
12	Exception, Error, Exception Handling, try-catch-finally block	CO3
13	Packages, Enums, File Handling	CO3
14	Coupling, Cohesion, Good vs bad design examples	CO4
15	Review	CO1, CO2, CO3, CO4
Semester Final Examinations		

Program Outcomes	
PO 1	**Engineering Knowledge:** Apply knowledge of **mathematics, natural science**, **engineering** fundamentals and system fundamentals, software development, networking & communication, and information assurance & security to the solution of complex engineering problems in computer science and engineering.
PO 2	**Problem Analysis:** Ability to **identify**, **formulate** and **analyze complex** Computer Science and Engineering problems in the areas of hardware, software, theoretical Computer Science and applications to reach significant conclusions by applying Mathematics, Natural sciences, Computer Science and Engineering principles.
PO 3	**Design/ Development of Solutions:** **Design solutions** for complex computer science and engineering problems and design systems, components or processes that meet specified needs with appropriate consideration for public health and safety, cultural, societal, and environmental considerations.
PO 4	**Investigation:** Ability to use **research based knowledge** and **research methods** to perform literature survey, design experiments for complex problems in designing, developing and maintaining a computing system, collect data from the experimental outcome, analyze and interpret valid/interesting patterns and conclusions from the data points.
PO 5	**Modern Tool Usage:** Ability to create, select and apply **state of the art tools** and techniques in designing, developing and testing a computing system or its components.
PO 6	**The Engineer and Society:** Apply reasoning informed by contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to **professional engineering practice** in system development and solutions to **complex engineering problems** related to system fundamentals, software development, networking & communication, and information assurance & security.
PO 7	**Environment and Sustainability:** Apply reasoning informed by contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to **professional engineering practice** in system development and solutions to **complex engineering problems** related to system fundamentals, software development, networking & communication, and information assurance & security.
PO 8	**Ethics:** Apply **ethical principles** and commit to **professional ethics** and **responsibilities** and norms of computer science and engineering practice.
PO 9	**Individual Work and Teamwork:** Ability to function as an individual and as a team player or leader in multidisciplinary teams and strive towards **achieving a common goal.**
PO 10	**Communication:** **Communicate effectively** on complex engineering activities with the engineering community and with society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.
PO 11	**Project Management and Finance:**

	Demonstrate knowledge and understanding of engineering management principles and economic decision making and apply these to one's own work, as a member and leader in a team, to **manage projects** and in multidisciplinary environments.
PO 12	**Life-long learning:** Recognize the need for, and have the preparation and ability to **engage in independent** and **lifelong learning** in the broadest context of technological change.

Mapping of COs and POs [Correlation level 1 for low, 2 for moderate and 3 for high]												
Course Outcomes	**PO1**	**PO2**	**PO3**	**PO4**	**PO5**	**PO6**	**PO7**	**PO8**	**PO9**	**PO10**	**PO11**	**PO12**
CO1	√											
CO2		√										
CO3			√		√							
CO4				√								

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